TCS NQT ADVANCED QUANTITATIVE PREP

Quantitative Aptitude: Alligations & Mixtures Masterclass

Weighted Balance Frameworks, Cross-Alligation Rules, and Milk-Water Replacement Rules

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Section 1: Core Concepts, Frameworks & Speed Shortcuts

Alligations and Mixtures represent one of the highest-yielding topics in the TCS NQT arithmetic section [cite: 504]. Rather than testing basic factual plug-and-play math, TCS designs multi-tiered mixture questions that require tracking percentage concentrations, relative values, or dynamic fluid displacement chains [cite: 2263]. Using algebraic equations with multiple variables (x, y) is highly inefficient under exam conditions. Mastering the shortcuts below will help you bypass fractional arithmetic and build absolute conceptual speed.

1. The Fundamental Cross-Alligation Rule

The rule of alligation is a mathematical shortcut used to find the ratio in which two sub-components must be mixed to achieve a specific target average concentration value.

The Alligation Cross-Grid Architecture:

Let C be the cost/concentration of the cheaper component, D be the cost/concentration of the dearer (more expensive) component, and M be the final targeted Mean Concentration.
Arrange the variables in a cross-grid to calculate absolute diagonal differences:

Cheaper (C)   Dearer (D)
\   /
Mean Target (M)
/   \
($D - M$)   ($M - C$)

The required mixing ratio is given directly by:

rac{ ext{Quantity of Cheaper } (Q_C)}{ ext{Quantity of Dearer } (Q_D)} = rac{D - M}{M - C}

2. The Weighted Balance Shortcut

When you need to find the final concentration of a mixture directly without building a cross-grid, look at the problem as a weighted average calculation:

The Absolute Concentration Formula:

Final Concentration (C_f) = rac{V_1 \cdot P_1 + V_2 \cdot P_2}{V_1 + V_2}

Where *V_1, V_2* are the component volumes and *P_1, P_2* are their respective percentage purity or concentration values. Just like in general averages, substituting the simplified ratio of the volumes instead of their absolute values saves significant calculation time.

3. Dynamic Fluid Successive Replacement Theorem

A classic, high-priority question type in the TCS NQT involves starting with a container full of pure liquid (e.g., pure milk), withdrawing a fixed volume x, replacing it with water, and repeating this operation multiple times.

The Successive Replacement Formula:

Let V be the initial total volume of pure liquid in the container, and x be the volume extracted and replaced in each round. After executing this exact operation n consecutive times, the final volume of the original pure liquid remaining in the container is:

ext{Pure Liquid Remaining} = V \cdot \left(1 - rac{x}{V} ight)^n

• **The Concentration Fraction:** The ratio of the final pure liquid to the total container capacity is simply *(1 - rac{x}{V})^n*.

Section 2: The Examiner's Trap Door - Where TCS Catches You

TCS aptitude assessments are designed to test your conceptual accuracy under time pressure. Watch out for these four common logical traps when solving mixtures and alligations questions.

TRAP 1: MIXING PROFIT PERCENTAGES WITH RAW COSTS

You can only apply the alligation rule to components that share the exact same unit base. If a problem gives you the Cost Price of Component A and Component B, but states that the final mixture is sold at a *Profit of 20%* for Rs. 60, you **cannot** use 60 as your mean value (M). Doing so will skew your diagonal differences.
The Fix: You must calculate the Cost Price of the mixture first ($CP = 60 / 1.20 = 50$) and use that cost value as your true mean center.

TRAP 2: THE MILK-TO-WATER VS. WATER-TO-MILK RATIO SHIFT

After calculating ratio components correctly (e.g., Cheaper = 3 parts, Dearer = 5 parts), candidates often rush and pick Option A, only to realize later that the question asked for the ratio of *Water to Milk* rather than *Milk to Water*. Always verify the precise phrasing of the required final ratio before submitting your answer.

TRAP 3: VARIABLE CONCENTRATIONS ACROSS MULTIPLE CONTAINERS

When mixing liquids from two separate vessels—say, Vessel 1 with a milk-to-water ratio of $5:3$ [cite: 2263] and Vessel 2 with a ratio of $7:1$—students often incorrectly add the ratio terms directly ($5+7 : 3+1$).
The Correct Base Method: Alligation must be performed using a single, consistent reference asset—such as the fractional concentration of milk ($ rac{5}{8}$ for Vessel 1 and $ rac{7}{8}$ for Vessel 2).

TRAP 4: FRACTIONAL TRAILING IN REPLACEMENTS ("WITHDRAWN FROM THE REMAINING")

In replacement problems, check the wording carefully to see if a fixed volume is withdrawn in each step, or if the problem says: *"1/4 of the fluid is removed and replaced with water, then 1/3 of the **remaining** mixture is removed..."*. If the volume changes between steps, you cannot use a simple exponent (n). Instead, multiply the remaining fractions sequentially: $V \cdot (1 - f_1) \cdot (1 - f_2)$.

Section 3: 10 High-Priority Practice Questions & Detailed Explanations

Question 1: Volumetric Pure Compound Water Addition Matrix [cite: 2263]

A laboratory reservoir contains a chemical mixture of pure milk and water structured in the ratio $5 : 3$ [cite: 2263]. When exactly 8 liters of pure water are added to the reservoir, the internal balance shifts, and the updated ratio becomes $5 : 5$ [cite: 2263]. Find the total initial quantity (in liters) of the mixture inside the reservoir [cite: 2264].

A) 20 liters [cite: 2266]

B) 32 liters ✓ [cite: 2267]

C) 40 liters [cite: 2268]

D) 48 liters [cite: 2269]

SOLUTION EXPLANATION:

This matches Question 248 from Section 28 of your reference material [cite: 2263]. Let's solve it efficiently by tracking the unchanged component.

• Initial ratio: $ext{Milk} : ext{Water} = 5 : 3$ [cite: 2263]. Let $ext{Milk} = 5x$ and $ext{Water} = 3x$ [cite: 2270].

• New ratio after adding water: $5 : 5$ [cite: 2263].

Notice that the milk component remains at 5 parts in both ratios. Since no milk was added or removed, we can track the volume change using the water parts directly:

• Increase in water parts = $5 ext{ parts} - 3 ext{ parts} = 2 ext{ parts}$.

The problem states this increase equals 8 liters [cite: 2263]:

$2 ext{ parts} = 8 ext{ liters} \rightarrow 1 ext{ part} = 4 ext{ liters}$ [cite: 2270].

Now, calculate the total initial volume of the mixture (which consisted of $5 + 3 = 8$ parts) [cite: 2263, 2270]:

$ext{Total Initial Volume} = 8 ext{ parts} \times 4 ext{ liters/part} = 32 ext{ liters}$ [cite: 2267, 2270].

Correct Answer: B [cite: 2267]

Question 2: Mean Cost-Price Adjustments with Profit Traps

A merchant mixes premium Basmati rice costing Rs. 60 per kg with an economy variety costing Rs. 45 per kg. He then sells the combined mixture at Rs. 66 per kg, generating a net profit of 20%. Find the ratio in which the two rice varieties were mixed.

A) 3 : 2

B) 1 : 2 ✓

C) 2 : 3

D) 2 : 1

SOLUTION EXPLANATION:

This targets Trap 1. Do not use the Selling Price of Rs. 66 directly in your alligation grid. First, calculate the true Cost Price of the mixture:

• *ext{Selling Price} = ext{Cost Price} × (1 + ext{Profit}\%) → 66 = ext{Cost Price} × 1.20*.

• *ext{True Mixture Cost Price } (M) = 66 / 1.20 = ext{Rs. } 55 ext{ per kg}*.

Now, set up your cross-alligation grid using this cost price base:

• Cheaper variety cost (C) = 45; Premium variety cost (D) = 60; Mixture mean cost (M) = 55.

Calculate the diagonal absolute differences:

• Right side variance ($M - C$) = $55 - 45 = 10$.

• Left side variance ($D - M$) = $60 - 55 = 5$.

The required ratio of cheaper to premium rice is:

ext{Ratio} = 5 : 10 = 1 : 2.

Correct Answer: B

Question 3: Chained Multi-Stage Successive Replacement Analysis

A barrel is initially filled with 80 liters of pure liquid chemical. An automated system extracts 8 liters of the chemical from the barrel and replaces it with pure water. This replacement operation is performed three times in total. Find the final volume of the original chemical remaining in the barrel.

A) 58.32 liters ✓

B) 56.00 liters

C) 60.24 liters

D) 54.58 liters

SOLUTION EXPLANATION:

Apply our Successive Replacement Formula here:

• Initial total volume (V) = 80 liters.

• Extracted volume per step (x) = 8 liters.

• Total number of cycles (n) = 3. *ext{Chemical Remaining} = 80 \cdot \left(1 - rac{8}{80} ight)^3 = 80 \cdot \left(1 - rac{1}{10} ight)^3 = 80 \cdot \left(rac{9}{10} ight)^3*

Calculate the cube of the fraction:

\left(rac{9}{10} ight)^3 = rac{729}{1000} = 0.729.

Now, multiply by the initial volume:

ext{Chemical Remaining} = 80 imes 0.729 = 58.32 ext{ liters}.

Correct Answer: A

Question 4: Dual-Vessel Concentration Convergence

Vessel X contains a solution of milk and water mixed in the ratio 4:3. Vessel Y contains a similar solution mixed in the ratio 2:3. In what ratio must the liquids be drawn from both vessels and mixed together to obtain a new solution with a milk-to-water ratio of exactly 1:1?

A) 7 : 5 ✓

B) 5 : 7

C) 3 : 2

D) 4 : 5

SOLUTION EXPLANATION:

This targets Trap 3. To solve this problem accurately, convert the mixtures into fractional concentrations using milk as your reference asset:

• Concentration of milk in Vessel X (C) = $4 / (4 + 3) = 4 / 7$.

• Concentration of milk in Vessel Y (D) = $2 / (2 + 3) = 2 / 5$.

• Target concentration of milk in the final mixture (M) = $1 / (1 + 1) = 1 / 2$.

Set up your cross-alligation grid using these fractions:

• Left side difference ($|2/5 - 1/2|$) = $|4/10 - 5/10| = 1 / 10$.

• Right side difference ($|4/7 - 1/2|$) = $|8/14 - 7/14| = 1 / 14$.

The required mixing ratio is the ratio of these differences:

ext{Ratio} = (1 / 10) : (1 / 14) = 14 : 10 = 7 : 5.

Correct Answer: A

Question 5: Pure Liquid Replacement Ratio Inversion

A container is initially filled with pure wine. A sommelier removes 20% of the wine and replaces it with water. He performs this operation twice in total. Find the final ratio of wine to water inside the container.

A) 16 : 9 ✓

B) 16 : 25

C) 9 : 16

D) 25 : 16

SOLUTION EXPLANATION:

Let's use our concentration fraction rule to simplify the problem.

• Initial concentration of pure wine = 100% (or 1).

• The replacement fraction in each step is 20% ($x/V = 0.20$). This leaves $1 - 0.20 = 0.80$ (or $4/5$) of the wine remaining after each round.

After repeating the operation twice ($n = 2$):

$\text{Fraction of Wine Remaining} = (0.80)^2 = 0.64 = 64 / 100 = 16 / 25$. The remaining volume must be water:

• $\text{Volume of Water} = 25 \text{ parts} - 16 \text{ parts} = 9 \text{ parts}$.

Therefore, the final ratio of wine to water is:

$\text{Ratio} = 16 : 9$. Watch out for Trap 2 (Option B lists 16:25, which represents wine to *total volume*, not wine to *water*)!

Correct Answer: A

Question 6: Three-Group Compound Percentage Alligation Matrix

An alloys lab melts down three distinct grades of copper wire. Grade A contains 60% pure copper, Grade B contains 70% pure copper, and Grade C contains 80% pure copper. If the lab mixes these components in the weight ratio of 1:2:3 respectively, what is the percentage concentration of pure copper in the resulting combined alloy?

A) 73.33% ✓

B) 70.00%

C) 75.50%

D) 72.00%

SOLUTION EXPLANATION:

Apply our Weighted Balance shortcut directly for a three-component system:

• Component weights: $w_1 = 1,\ w_2 = 2,\ w_3 = 3$. Total combined weight = $1 + 2 + w_3 = 6 \text{ parts}$.

• Concentration values: $P_1 = 60\%,\ P_2 = 70\%,\ P_3 = 80\%$.

Calculate the final weighted average concentration:

$$P_f = \frac{(1 \times 60) + (2 \times 70) + (3 \times 80)}{1 + 2 + 3}$$

$P_f = (60 + 140 + 240) / 6 = 440 / 6 = 220 / 3 \approx 73.33\%$.

The resulting alloy has a pure copper concentration of exactly 73.33%.

Correct Answer: A

Question 7: Profit-Driven Acid Alligation Models

A chemical supplier sells a concentrated acid mixture at Rs. 23 per liter, making a clean profit of 15% on the sale. If this mixture was produced by combining pure acid costing Rs. 25 per liter with free water, find the ratio of acid to water in the supplier's product.

A) 4 : 1 ✓

B) 5 : 1

C) 3 : 1

D) 4 : 3

SOLUTION EXPLANATION:

First, clear the profit trap by finding the true base Cost Price of the mixture:

• *ext{Cost Price} = ext{Selling Price} / (1 + ext{Profit}\%) = 23 / 1.15 = ext{Rs. } 20 ext{ per liter}*.

Now, set up your cross-alligation grid using this cost price base. Note that the problem treats water as free, so its cost is Rs. 0:

• Water cost (C) = 0; Pure acid cost (D) = 25; Mixture mean cost (M) = 20.

Calculate the diagonal absolute differences:

• Left variance ($D - M$) = *25 - 20 = 5*.

• Right variance ($M - C$) = *20 - 0 = 20*.

The required ratio of pure acid to water is:

ext{Ratio} = 20 : 5 = 4 : 1.

Correct Answer: A

Question 8: Fractional Volumetric Extraction Dynamics

A vat is filled with 60 liters of pure organic juice. A technician removes 15 liters of juice from the vat and replaces it with water. Next, he removes 20 liters of this mixture and replaces it with water. Find the final volume of pure juice remaining in the vat.

A) 30 liters ✓

B) 35 liters

C) 25 liters

D) 32 liters

SOLUTION EXPLANATION:

This targets Trap 4. Since the volume extracted changes between steps, compute the remaining juice fractions sequentially instead of using a single exponent:

• **Step 1:** 15 liters are removed from the 60-liter total. Extracted fraction $f_1 = 15/60 = 1/4$, leaving $1 - 1/4 = 3/4$ of the juice remaining.

• **Step 2:** 20 liters are removed from the 60-liter total mixture. Extracted fraction $f_2 = 20/60 = 1/3$, leaving $1 - 1/3 = 2/3$ of the juice remaining.

Now, calculate the final volume of pure juice by multiplying these remaining fractions together:

ext{Final Juice Volume} = 60 imes \left(rac{3}{4} ight) imes \left(rac{2}{3} ight) = 60 imes \left(rac{6}{12} ight) = 60 imes \left(rac{1}{2} ight) = 30 ext{ liters}.

The vat contains exactly 30 liters of pure juice at the end of the operations.

Correct Answer: A

Question 9: Reverse Alligation Capital-Volume Solving

A dairy farmer mixes pure milk costing Rs. 50 per liter with water to produce a commercial batch worth Rs. 40 per liter. If the total volume of the resulting batch is exactly 150 liters, how many liters of water did the farmer add?

A) 30 liters ✓ B) 25 liters C) 40 liters D) 35 liters

SOLUTION EXPLANATION:

Set up your cross-alligation grid using water (cost = Rs. 0) and pure milk (cost = Rs. 50) to reach the target mean value of Rs. 40:

• Water cost (C) = 0; Milk cost (D) = 50; Target mean cost (M) = 40.

Calculate the diagonal absolute differences:

• Left side difference ($D - M$) = $50 - 40 = 10$.

• Right side difference ($M - C$) = $40 - 0 = 40$.

The required ratio of water to milk is:

$ext{Ratio} = 10 : 40 = 1 : 4$.

Divide the total volume of 150 liters into $1 + 4 = 5$ proportional parts:

• $5 ext{ parts} = 150 ext{ liters} \rightarrow 1 ext{ part} = 150 / 5 = 30 ext{ liters}$.

Since water corresponds to exactly 1 part in our ratio, the farmer added 30 liters of water.

Correct Answer: A

Question 10: Percentage Difference Boundary Intersections

Vessel A contains an alcohol solution with a concentration of 25%, while Vessel B contains a stronger solution with a concentration of 40%. A lab tech draws liquid from both vessels to create a new 60-liter solution with an alcohol concentration of 30%. Find the volume of liquid drawn from Vessel A.

A) 40 liters ✓

B) 20 liters

C) 30 liters

D) 45 liters

SOLUTION EXPLANATION:

Set up your cross-alligation grid using the individual percentage concentrations directly:

• Concentration A (C) = 25%; Concentration B (D) = 40%; Target mean concentration (M) = 30%.

Calculate the diagonal absolute differences:

• Left side difference ($D - M$) = $40 - 30 = 10$.

• Right side difference ($M - C$) = $30 - 25 = 5$.

The ratio in which the solutions must be mixed is:

$ext{Ratio (A : B)} = 10 : 5 = 2 : 1$.

Divide the total target volume of 60 liters into $2 + 1 = 3$ proportional parts:

• $3 ext{ parts} = 60 ext{ liters} \rightarrow 1 ext{ part} = 60 / 3 = 20 ext{ liters}$.

The volume drawn from Vessel A corresponds to 2 parts:

• $ext{Volume A} = 2 ext{ parts} \times 20 ext{ liters/part} = 40 ext{ liters}$.

Correct Answer: A